$C_{qd}^{(8)}$ i1_i2_i3_i4_ →

$$\frac{1}{16\pi^2}\Big(\frac{2}{3} g_3^4 \mathrm{LF}_{3,0}[m_3]\,\delta_{i1i2}\delta_{i3i4} + g_3^4 \mathrm{LF}_{4,-1}[m_3]\,\delta_{i1i2}\delta_{i3i4} - \frac{16}{15} g_3^4 \mathrm{LF}_{5,-2}[m_3]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{2}{9}\sum_p g_3^4 \mathrm{LF}_{3,0}[m_{\tilde{d}}^{p}]\,\delta_{i1i2}\delta_{i3i4} - \frac{5}{12}\sum_p g_3^4 \mathrm{LF}_{4,-1}[m_{\tilde{d}}^{p}]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{8}{45}\sum_p g_3^4 \mathrm{LF}_{5,-2}[m_{\tilde{d}}^{p}]\,\delta_{i1i2}\delta_{i3i4} + \frac{4}{9}\sum_p g_3^4 \mathrm{LF}_{3,0}[m_{\tilde{q}}^{p}]\,\delta_{i1i2}\delta_{i3i4} -$$

$$\frac{5}{6}\sum_p g_3^4 \mathrm{LF}_{4,-1}[m_{\tilde{q}}^{p}]\,\delta_{i1i2}\delta_{i3i4} + \frac{16}{45}\sum_p g_3^4 \mathrm{LF}_{5,-2}[m_{\tilde{q}}^{p}]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{2}{9}\sum_p g_3^4 \mathrm{LF}_{3,0}[m_{\tilde{u}}^{p}]\,\delta_{i1i2}\delta_{i3i4} - \frac{5}{12}\sum_p g_3^4 \mathrm{LF}_{4,-1}[m_{\tilde{u}}^{p}]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{8}{45}\sum_p g_3^4 \mathrm{LF}_{5,-2}[m_{\tilde{u}}^{p}]\,\delta_{i1i2}\delta_{i3i4} + \frac{1}{54} g_1^2 g_3^2 \mathrm{LF}_{2,1,0}[m_1, m_{\tilde{d}}^{i3}]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{54} g_1^2 g_3^2 \mathrm{LF}_{2,2,-1}[m_1, m_{\tilde{d}}^{i3}]\,\delta_{i1i2}\delta_{i3i4} - \frac{1}{27} g_1^2 g_3^2 \mathrm{LF}_{3,1,-1}[m_1, m_{\tilde{d}}^{i3}]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{54} g_1^2 g_3^2 \mathrm{LF}_{4,1,-2}[m_1, m_{\tilde{d}}^{i3}]\,\delta_{i1i2}\delta_{i3i4} + \frac{1}{54} g_1^2 g_3^2 \mathrm{LF}_{2,1,0}[m_1, m_{\tilde{d}}^{i4}]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{54} g_1^2 g_3^2 \mathrm{LF}_{2,2,-1}[m_1, m_{\tilde{d}}^{i4}]\,\delta_{i1i2}\delta_{i3i4} - \frac{1}{27} g_1^2 g_3^2 \mathrm{LF}_{3,1,-1}[m_1, m_{\tilde{d}}^{i4}]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{54} g_1^2 g_3^2 \mathrm{LF}_{4,1,-2}[m_1, m_{\tilde{d}}^{i4}]\,\delta_{i1i2}\delta_{i3i4} + \frac{1}{216} g_1^2 g_3^2 \mathrm{LF}_{2,1,0}[m_1, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{216} g_1^2 g_3^2 \mathrm{LF}_{2,2,-1}[m_1, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\delta_{i3i4} - \frac{1}{108} g_1^2 g_3^2 \mathrm{LF}_{3,1,-1}[m_1, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{216} g_1^2 g_3^2 \mathrm{LF}_{4,1,-2}[m_1, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\delta_{i3i4} + \frac{1}{216} g_1^2 g_3^2 \mathrm{LF}_{2,1,0}[m_1, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{216} g_1^2 g_3^2 \mathrm{LF}_{2,2,-1}[m_1, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\delta_{i3i4} - \frac{1}{108} g_1^2 g_3^2 \mathrm{LF}_{3,1,-1}[m_1, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{216} g_1^2 g_3^2 \mathrm{LF}_{4,1,-2}[m_1, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\delta_{i3i4} + \frac{1}{8} g_2^2 g_3^2 \mathrm{LF}_{2,1,0}[m_2, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{8} g_2^2 g_3^2 \mathrm{LF}_{2,2,-1}[m_2, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\delta_{i3i4} - \frac{1}{4} g_2^2 g_3^2 \mathrm{LF}_{3,1,-1}[m_2, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{8} g_2^2 g_3^2 \mathrm{LF}_{4,1,-2}[m_2, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\delta_{i3i4} + \frac{1}{8} g_2^2 g_3^2 \mathrm{LF}_{2,1,0}[m_2, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{8} g_2^2 g_3^2 \mathrm{LF}_{2,2,-1}[m_2, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\delta_{i3i4} - \frac{1}{4} g_2^2 g_3^2 \mathrm{LF}_{3,1,-1}[m_2, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{8} g_2^2 g_3^2 \mathrm{LF}_{4,1,-2}[m_2, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\delta_{i3i4} - \frac{1}{36} g_3^4 \mathrm{LF}_{2,1,0}[m_3, m_{\tilde{d}}^{i3}]\,\delta_{i1i2}\delta_{i3i4} -$$

$$\frac{1}{36} g_3^4 \mathrm{LF}_{2,2,-1}[m_3, m_{\tilde{d}}^{i3}]\,\delta_{i1i2}\delta_{i3i4} - \frac{25}{36} g_3^4 \mathrm{LF}_{3,1,-1}[m_3, m_{\tilde{d}}^{i3}]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{2}{9} g_3^4 \mathrm{LF}_{4,1,-2}[m_3, m_{\tilde{d}}^{i3}]\,\delta_{i1i2}\delta_{i3i4} - \frac{1}{36} g_3^4 \mathrm{LF}_{2,1,0}[m_3, m_{\tilde{d}}^{i4}]\,\delta_{i1i2}\delta_{i3i4} -$$

$$\frac{1}{36} g_3^4 \mathrm{LF}_{2,2,-1}[m_3, m_{\tilde{d}}^{i4}]\,\delta_{i1i2}\delta_{i3i4} - \frac{25}{36} g_3^4 \mathrm{LF}_{3,1,-1}[m_3, m_{\tilde{d}}^{i4}]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{2}{9} g_3^4 \mathrm{LF}_{4,1,-2}[m_3, m_{\tilde{d}}^{i4}]\,\delta_{i1i2}\delta_{i3i4} - \frac{1}{36} g_3^4 \mathrm{LF}_{2,1,0}[m_3, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\delta_{i3i4} -$$

$$\frac{1}{36} g_3^4 \mathrm{LF}_{2,2,-1}[m_3, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\delta_{i3i4} - \frac{25}{36} g_3^4 \mathrm{LF}_{3,1,-1}[m_3, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{2}{9} g_3^4 \mathrm{LF}_{4,1,-2}[m_3, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\delta_{i3i4} - \frac{1}{36} g_3^4 \mathrm{LF}_{2,1,0}[m_3, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\delta_{i3i4} -$$

$$\frac{1}{36} g_3^4 \mathrm{LF}_{2,2,-1}[m_3, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\delta_{i3i4} - \frac{25}{36} g_3^4 \mathrm{LF}_{3,1,-1}[m_3, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{2}{9} g_3^4 \mathrm{LF}_{4,1,-2}[m_3, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\delta_{i3i4} - \frac{1}{3} g_3^2\, \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{2,1,0}[m_{\tilde{d}}^{p}, \tilde{\mu}]\,\delta_{i3i4} +$$

$$\frac{1}{6} g_3^2\, \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{2,2,-1}[m_{\tilde{d}}^{p}, \tilde{\mu}]\,\delta_{i3i4} + \frac{1}{6} g_3^2\, \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{3,1,-1}[m_{\tilde{d}}^{p}, \tilde{\mu}]\,\delta_{i3i4} -$$

$$\frac{1}{27} g_1^2 g_3^2 \mathrm{LF}_{2,1,0}[m_{\tilde{d}}^{i3}, m_1]\,\delta_{i1i2}\delta_{i3i4} + \frac{1}{54} g_1^2 g_3^2 \mathrm{LF}_{3,1,-1}[m_{\tilde{d}}^{i3}, m_1]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{18} g_3^4 \mathrm{LF}_{2,1,0}[m_{\tilde{d}}^{i3}, m_3]\,\delta_{i1i2}\delta_{i3i4} - \frac{1}{36} g_3^4 \mathrm{LF}_{3,1,-1}[m_{\tilde{d}}^{i3}, m_3]\,\delta_{i1i2}\delta_{i3i4} -$$

$$\frac{1}{27} g_1^2 g_3^2 \mathrm{LF}_{2,1,0}[m_{\tilde{d}}^{i4}, m_1]\,\delta_{i1i2}\delta_{i3i4} + \frac{1}{54} g_1^2 g_3^2 \mathrm{LF}_{3,1,-1}[m_{\tilde{d}}^{i4}, m_1]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{18} g_3^4 \mathrm{LF}_{2,1,0}[m_{\tilde{d}}^{i4}, m_3]\,\delta_{i1i2}\delta_{i3i4} - \frac{1}{36} g_3^4 \mathrm{LF}_{3,1,-1}[m_{\tilde{d}}^{i4}, m_3]\,\delta_{i1i2}\delta_{i3i4} -$$

$$\overline{y_d}^{i2i3} y_d^{i1i4}\, \overline{a_e}^{pr} a_e^{pr} \mathrm{LF}_{3,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] + 3\, \overline{y_d}^{i2i3} y_d^{i1i4}\, \overline{a_e}^{pr} a_e^{pr} \mathrm{LF}_{4,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] -$$

$$2\, \overline{y_d}^{i2i3} y_d^{i1i4}\, \overline{a_e}^{pr} a_e^{pr} \mathrm{LF}_{5,1,-2}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] - 3\, \overline{y_d}^{i2i3} y_d^{i1i4}\, \overline{a_d}^{pr} a_d^{pr} \mathrm{LF}_{3,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] +$$

$$9\, \overline{y_d}^{i2i3} y_d^{i1i4}\, \overline{a_d}^{pr} a_d^{pr} \mathrm{LF}_{4,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] - 6\, \overline{y_d}^{i2i3} y_d^{i1i4}\, \overline{a_d}^{pr} a_d^{pr} \mathrm{LF}_{5,1,-2}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] -$$

$$\frac{2}{3} g_3^2\, \overline{y_d}^{pi3} y_d^{pi4} \mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{p}, \tilde{\mu}]\,\delta_{i1i2} + \frac{1}{3} g_3^2\, \overline{y_d}^{pi3} y_d^{pi4} \mathrm{LF}_{2,2,-1}[m_{\tilde{q}}^{p}, \tilde{\mu}]\,\delta_{i1i2} +$$

$$\frac{1}{3} g_3^2\, \overline{y_d}^{pi3} y_d^{pi4} \mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{p}, \tilde{\mu}]\,\delta_{i1i2} - \frac{1}{108} g_1^2 g_3^2 \mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i1}, m_1]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{216} g_1^2 g_3^2 \mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i1}, m_1]\,\delta_{i1i2}\delta_{i3i4} - \frac{1}{4} g_2^2 g_3^2 \mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i1}, m_2]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{8} g_2^2 g_3^2 \mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i1}, m_2]\,\delta_{i1i2}\delta_{i3i4} + \frac{1}{18} g_3^4 \mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i1}, m_3]\,\delta_{i1i2}\delta_{i3i4} -$$

$$\frac{1}{36} g_3^4 \mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i1}, m_3]\,\delta_{i1i2}\delta_{i3i4} - \frac{1}{108} g_1^2 g_3^2 \mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i2}, m_1]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{216} g_1^2 g_3^2 \mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i2}, m_1]\,\delta_{i1i2}\delta_{i3i4} - \frac{1}{4} g_2^2 g_3^2 \mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i2}, m_2]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{8} g_2^2 g_3^2 \mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i2}, m_2]\,\delta_{i1i2}\delta_{i3i4} + \frac{1}{18} g_3^4 \mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i2}, m_3]\,\delta_{i1i2}\delta_{i3i4} -$$

$$\frac{1}{36} g_3^4 \mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i2}, m_3]\,\delta_{i1i2}\delta_{i3i4} - \frac{1}{3} g_3^2\, \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{2,1,0}[m_{\tilde{u}}^{p}, \tilde{\mu}]\,\delta_{i3i4} +$$

$$\frac{1}{6} g_3^2\, \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{2,2,-1}[m_{\tilde{u}}^{p}, \tilde{\mu}]\,\delta_{i3i4} + \frac{1}{6} g_3^2\, \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{3,1,-1}[m_{\tilde{u}}^{p}, \tilde{\mu}]\,\delta_{i3i4} -$$

$$3\tilde{\mu}^2\, \overline{y_d}^{i2i3} y_d^{i1i4}\, \overline{y_u}^{pr} y_u^{pr} \mathrm{LF}_{3,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] + 9\tilde{\mu}^2\, \overline{y_d}^{i2i3} y_d^{i1i4}\, \overline{y_u}^{pr} y_u^{pr} \mathrm{LF}_{4,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] -$$

$$6\tilde{\mu}^2\, \overline{y_d}^{i2i3} y_d^{i1i4}\, \overline{y_u}^{pr} y_u^{pr} \mathrm{LF}_{5,1,-2}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] + 2 g_1^2\, \overline{y_d}^{i2i3} y_d^{i1i4} \mathrm{LF}_{3,1,-1}[\tilde{\mu}, m_1] -$$

$$4 g_1^2\, \overline{y_d}^{i2i3} y_d^{i1i4} \mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_1] + 2 g_1^2\, \overline{y_d}^{i2i3} y_d^{i1i4} \mathrm{LF}_{5,1,-3}[\tilde{\mu}, m_1] +$$

$$6 g_2^2\, \overline{y_d}^{i2i3} y_d^{i1i4} \mathrm{LF}_{3,1,-1}[\tilde{\mu}, m_2] - 12 g_2^2\, \overline{y_d}^{i2i3} y_d^{i1i4} \mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_2] +$$

$$6 g_2^2\, \overline{y_d}^{i2i3} y_d^{i1i4} \mathrm{LF}_{5,1,-3}[\tilde{\mu}, m_2] + \frac{1}{6} g_3^2\, \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{2,1,0}[\tilde{\mu}, m_{\tilde{d}}^{p}]\,\delta_{i3i4} -$$

$$\frac{1}{3} g_3^2\, \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{3,1,-1}[\tilde{\mu}, m_{\tilde{d}}^{p}]\,\delta_{i3i4} + \frac{1}{6} g_3^2\, \overline{y_d}^{i2p} y_d^{i1p} \mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_{\tilde{d}}^{p}]\,\delta_{i3i4} +$$

$$\frac{1}{3} g_3^2\, \overline{y_d}^{pi3} y_d^{pi4} \mathrm{LF}_{2,1,0}[\tilde{\mu}, m_{\tilde{q}}^{p}]\,\delta_{i1i2} - \frac{2}{3} g_3^2\, \overline{y_d}^{pi3} y_d^{pi4} \mathrm{LF}_{3,1,-1}[\tilde{\mu}, m_{\tilde{q}}^{p}]\,\delta_{i1i2} +$$

$$\frac{1}{3} g_3^2\, \overline{y_d}^{pi3} y_d^{pi4} \mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_{\tilde{q}}^{p}]\,\delta_{i1i2} + \frac{1}{6} g_3^2\, \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{2,1,0}[\tilde{\mu}, m_{\tilde{u}}^{p}]\,\delta_{i3i4} -$$

$$\frac{1}{3} g_3^2\, \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{3,1,-1}[\tilde{\mu}, m_{\tilde{u}}^{p}]\,\delta_{i3i4} + \frac{1}{6} g_3^2\, \overline{y_u}^{i2p} y_u^{i1p} \mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_{\tilde{u}}^{p}]\,\delta_{i3i4} -$$

$$\frac{1}{36} m_1 g_1^2 y_d^{i1i4}\, \overline{a_d}^{i2i3} \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{d}}^{i3}, m_{\tilde{q}}^{i2}] - \frac{1}{36} m_1 g_1^2\, \overline{y_d}^{i2i3} a_d^{i1i4}$$

$$\mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{d}}^{i4}, m_{\tilde{q}}^{i1}] - \frac{1}{36} m_1 g_1^2\, \overline{y_d}^{i2i3} a_d^{i1i4} \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{d}}^{i4}] -$$

$$\frac{1}{36} m_1 g_1^2 y_d^{i1i4}\, \overline{a_d}^{i2i3} \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{q}}^{i2}, m_{\tilde{d}}^{i3}] - \frac{1}{6} g_1^2\, \overline{y_d}^{i2i3} y_d^{i1i4} \mathrm{LF}_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{d}}^{i3}] -$$

$$\frac{1}{6} g_1^2\, \overline{y_d}^{i2i3} y_d^{i1i4} \mathrm{LF}_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{d}}^{i4}] - \frac{1}{12} g_1^2\, \overline{y_d}^{i2i3} y_d^{i1i4} \mathrm{LF}_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{q}}^{i1}] -$$

$$\frac{1}{12} g_1^2\, \overline{y_d}^{i2i3} y_d^{i1i4} \mathrm{LF}_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{q}}^{i2}] - \frac{3}{4} g_2^2\, \overline{y_d}^{i2i3} y_d^{i1i4} \mathrm{LF}_{2,2,1,-2}[m_2, \tilde{\mu}, m_{\tilde{q}}^{i1}] -$$

$$\frac{3}{4} g_2^2\, \overline{y_d}^{i2i3} y_d^{i1i4} \mathrm{LF}_{2,2,1,-2}[m_2, \tilde{\mu}, m_{\tilde{q}}^{i2}] + \frac{1}{6} g_3^4 \mathrm{LF}_{2,1,1,-1}[m_3, m_{\tilde{d}}^{i3}, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\delta_{i3i4} -$$

$$\frac{7}{6} g_3^4 m_3^2 \mathrm{LF}_{2,1,1,0}[m_3, m_{\tilde{d}}^{i3}, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\delta_{i3i4} + \frac{2}{3} m_3 g_3^2 y_d^{i1i4}\, \overline{a_d}^{i2i3} \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde{d}}^{i3}, m_{\tilde{q}}^{i2}] +$$

$$\frac{2}{3} m_3 g_3^2\, \overline{y_d}^{i2i3} a_d^{i1i4} \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde{d}}^{i4}, m_{\tilde{q}}^{i1}] + \frac{1}{6} g_3^4 \mathrm{LF}_{2,1,1,-1}[m_3, m_{\tilde{d}}^{i4}, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\delta_{i3i4} -$$

$$\frac{7}{6} g_3^4 m_3^2 \mathrm{LF}_{2,1,1,0}[m_3, m_{\tilde{d}}^{i4}, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\delta_{i3i4} + \frac{2}{3} m_3 g_3^2\, \overline{y_d}^{i2i3} a_d^{i1i4} \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde{q}}^{i1}, m_{\tilde{d}}^{i4}] +$$

$$\frac{2}{3} m_3 g_3^2 y_d^{i1i4}\, \overline{a_d}^{i2i3} \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde{q}}^{i2}, m_{\tilde{d}}^{i3}] + \frac{1}{18} m_1 g_1^2 y_d^{i1i4}\, \overline{a_d}^{i2i3}$$

$$\mathrm{LF}_{2,1,1,0}[m_{\tilde{d}}^{i3}, m_1, m_{\tilde{q}}^{i2}] - \frac{1}{18} m_1 g_1^2 y_d^{i1i4}\, \overline{a_d}^{i2i3} \mathrm{LF}_{3,1,1,-1}[m_{\tilde{d}}^{i3}, m_1, m_{\tilde{q}}^{i2}] -$$

$$\frac{4}{3} m_3 g_3^2 y_d^{i1i4}\, \overline{a_d}^{i2i3} \mathrm{LF}_{2,1,1,0}[m_{\tilde{d}}^{i3}, m_3, m_{\tilde{q}}^{i2}] + \frac{4}{3} m_3 g_3^2 y_d^{i1i4}\, \overline{a_d}^{i2i3}$$

$$\mathrm{LF}_{3,1,1,-1}[m_{\tilde{d}}^{i3}, m_3, m_{\tilde{q}}^{i2}] + \frac{1}{18} m_1 g_1^2\, \overline{y_d}^{i2i3} a_d^{i1i4} \mathrm{LF}_{2,1,1,0}[m_{\tilde{d}}^{i4}, m_1, m_{\tilde{q}}^{i1}] -$$

$$\frac{1}{18} m_1 g_1^2\, \overline{y_d}^{i2i3} a_d^{i1i4} \mathrm{LF}_{3,1,1,-1}[m_{\tilde{d}}^{i4}, m_1, m_{\tilde{q}}^{i1}] - \frac{4}{3} m_3 g_3^2\, \overline{y_d}^{i2i3} a_d^{i1i4}$$

$$\mathrm{LF}_{2,1,1,0}[m_{\tilde{d}}^{i4}, m_3, m_{\tilde{q}}^{i1}] + \frac{4}{3} m_3 g_3^2\, \overline{y_d}^{i2i3} a_d^{i1i4} \mathrm{LF}_{3,1,1,-1}[m_{\tilde{d}}^{i4}, m_3, m_{\tilde{q}}^{i1}] -$$

$$\frac{1}{2}\tilde{\mu}^2 \left(\overline{y_d}^{pi3} y_d^{i1i4}\, \overline{y_u}^{i2r} y_u^{pr} + \overline{y_d}^{i2i3} y_d^{pi4}\, \overline{y_u}^{pr} y_u^{i1r}\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, \tilde{\mu}] +$$

$$\frac{1}{2}\tilde{\mu}^2 \left(\overline{y_d}^{pi3} y_d^{i1i4}\, \overline{y_u}^{i2r} y_u^{pr} + \overline{y_d}^{i2i3} y_d^{pi4}\, \overline{y_u}^{pr} y_u^{i1r}\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, \tilde{\mu}] +$$

$$\frac{1}{4}\tilde{\mu}^2 \left(\overline{y_d}^{pi3} y_d^{i1i4}\, \overline{y_u}^{i2r} y_u^{pr} + \overline{y_d}^{i2i3} y_d^{pi4}\, \overline{y_u}^{pr} y_u^{i1r}\right) \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{p}, \tilde{\mu}, m_{\tilde{u}}^{r}] +$$

$$\frac{1}{18} m_1 g_1^2\, \overline{y_d}^{i2i3} a_d^{i1i4} \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{i1}, m_1, m_{\tilde{d}}^{i4}] -$$

$$\frac{1}{18} m_1 g_1^2\, \overline{y_d}^{i2i3} a_d^{i1i4} \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{i1}, m_1, m_{\tilde{d}}^{i4}] -$$

$$\frac{4}{3} m_3 g_3^2\, \overline{y_d}^{i2i3} a_d^{i1i4} \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{i1}, m_3, m_{\tilde{d}}^{i4}] + \frac{4}{3} m_3 g_3^2\, \overline{y_d}^{i2i3} a_d^{i1i4}$$

$$\mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{i1}, m_3, m_{\tilde{d}}^{i4}] + \frac{1}{18} m_1 g_1^2 y_d^{i1i4}\, \overline{a_d}^{i2i3} \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{i2}, m_1, m_{\tilde{d}}^{i3}] -$$

$$\frac{1}{18} m_1 g_1^2 y_d^{i1i4}\, \overline{a_d}^{i2i3} \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{i2}, m_1, m_{\tilde{d}}^{i3}] - \frac{4}{3} m_3 g_3^2 y_d^{i1i4}\, \overline{a_d}^{i2i3}$$

$$\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{i2}, m_3, m_{\tilde{d}}^{i3}] + \frac{4}{3} m_3 g_3^2 y_d^{i1i4}\, \overline{a_d}^{i2i3} \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{i2}, m_3, m_{\tilde{d}}^{i3}] -$$

$$\frac{1}{2}\tilde{\mu}^2 \left(\overline{y_d}^{pi3} y_d^{i1i4}\, \overline{y_u}^{i2r} y_u^{pr} + \overline{y_d}^{i2i3} y_d^{pi4}\, \overline{y_u}^{pr} y_u^{i1r}\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, \tilde{\mu}] +$$

$$\frac{1}{2}\tilde{\mu}^2 \left(\overline{y_d}^{pi3} y_d^{i1i4}\, \overline{y_u}^{i2r} y_u^{pr} + \overline{y_d}^{i2i3} y_d^{pi4}\, \overline{y_u}^{pr} y_u^{i1r}\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, \tilde{\mu}] +$$

$$\frac{1}{4}\tilde{\mu}^2 \left(\overline{y_d}^{pi3} y_d^{i1i4}\, \overline{y_u}^{i2r} y_u^{pr} + \overline{y_d}^{i2i3} y_d^{pi4}\, \overline{y_u}^{pr} y_u^{i1r}\right) \mathrm{LF}_{2,2,1,-1}[m_{\tilde{u}}^{r}, \tilde{\mu}, m_{\tilde{q}}^{p}] +$$

$$\frac{1}{18} g_1^2 g_3^2 \mathrm{LF}_{1,1,1,1,-1}[m_1, m_3, m_{\tilde{d}}^{i3}, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{9} m_1 m_3 g_1^2 g_3^2 \mathrm{LF}_{1,1,1,1,0}[m_1, m_3, m_{\tilde{d}}^{i3}, m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{18} g_1^2 g_3^2 \mathrm{LF}_{1,1,1,1,-1}[m_1, m_3, m_{\tilde{d}}^{i4}, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{9} m_1 m_3 g_1^2 g_3^2 \mathrm{LF}_{1,1,1,1,0}[m_1, m_3, m_{\tilde{d}}^{i4}, m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{9} g_1^2\, \overline{y_d}^{i2i3} y_d^{i1i4} \mathrm{LF}_{1,1,1,1,-1}[m_1, m_{\tilde{d}}^{i3}, m_{\tilde{d}}^{i4}, \tilde{\mu}] + \frac{1}{18} g_1^2\, \overline{y_d}^{i2i3} y_d^{i1i4}$$

$$\mathrm{LF}_{1,1,1,1,-1}[m_1, m_{\tilde{d}}^{i3}, m_{\tilde{q}}^{i1}, \tilde{\mu}] + \frac{1}{18} g_1^2\, \overline{y_d}^{i2i3} y_d^{i1i4} \mathrm{LF}_{1,1,1,1,-1}[m_1, m_{\tilde{d}}^{i4}, m_{\tilde{q}}^{i2}, \tilde{\mu}] +$$

$$\frac{1}{36} g_1^2\, \overline{y_d}^{i2i3} y_d^{i1i4} \mathrm{LF}_{1,1,1,1,-1}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}, \tilde{\mu}] + \frac{3}{4} g_2^2\, \overline{y_d}^{i2i3} y_d^{i1i4}$$

$$\mathrm{LF}_{1,1,1,1,-1}[m_2, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}, \tilde{\mu}] - \frac{1}{6} g_3^2\, \overline{y_d}^{i2i3} y_d^{i1i4} \mathrm{LF}_{1,1,1,1,-1}[m_3, m_{\tilde{d}}^{i3}, m_{\tilde{d}}^{i4}, \tilde{\mu}] +$$

$$\frac{1}{6} g_3^2\, \overline{y_d}^{i2i3} y_d^{i1i4} \mathrm{LF}_{1,1,1,1,-1}[m_3, m_{\tilde{d}}^{i3}, m_{\tilde{q}}^{i1}, \tilde{\mu}] + \frac{1}{6} g_3^2\, \overline{y_d}^{i2i3} y_d^{i1i4}$$

$$\mathrm{LF}_{1,1,1,1,-1}[m_3, m_{\tilde{d}}^{i4}, m_{\tilde{q}}^{i2}, \tilde{\mu}] - \frac{1}{6} g_3^2\, \overline{y_d}^{i2i3} y_d^{i1i4} \mathrm{LF}_{1,1,1,1,-1}[m_3, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}, \tilde{\mu}]\Big)$$